CPE 4040: Data Collection and Analysis, Fall 2025

Final Project

Drowsiness Detection

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I. Objective

The objective of this final project was to design, implement, and evaluate a multimodal **driver drowsiness detection system** using real-time sensor data and machine-learning-based eye–state measurements. The system integrates:

- **EAR (Eye Aspect Ratio)** for blink detection and eyelid-openness tracking
- **PERCLOS (Percentage of Eye Closure)** for continuous fatigue estimation
- **Reaction time measurements** for performance-based alertness validation

Using collected alert and drowsy driving sessions, the goal was to analyze patterns in eye behavior, compute fatigue metrics, detect microsleep-like events, and determine how well these signals differentiate alert and drowsy conditions.

II. Material List

Hardware

- Laptop running Jupyter Notebook
- Camera for video-based eye tracking
- Sensor hardware used for reaction time measurement (keyboard/mouse or custom interface)

Software

- Python 3
- Jupyter Notebook
- OpenCV for video frame analysis
- pandas, NumPy, Matplotlib, seaborn for data analysis and visualization
- Custom Dataset class for structured data loading and processing
- CSV data files generated from alert/drowsy sessions:
    - EAR & PERCLOS files
    - Reaction time measurement files

III. Lab Procedures and Results

1. Overview of Approach

This project involved designing and evaluating a driver drowsiness detection system using video-based eye-tracking metrics (EAR and PERCLOS) and reaction-time measurements. The data collection, preprocessing, and analysis were carried out inside a Jupyter Notebook environment using Python and common scientific libraries.

The full workflow consisted of:

1. Collecting "alert" and "drowsy" driving-session data
2. Extracting EAR and PERCLOS values from each frame
3. Recording reaction-time responses during each condition
4. Loading and organizing the datasets using a custom `Dataset` class
5. Performing statistical analysis, event detection, and visualization
6. Comparing the two conditions to assess how eye behavior corresponds to fatigue

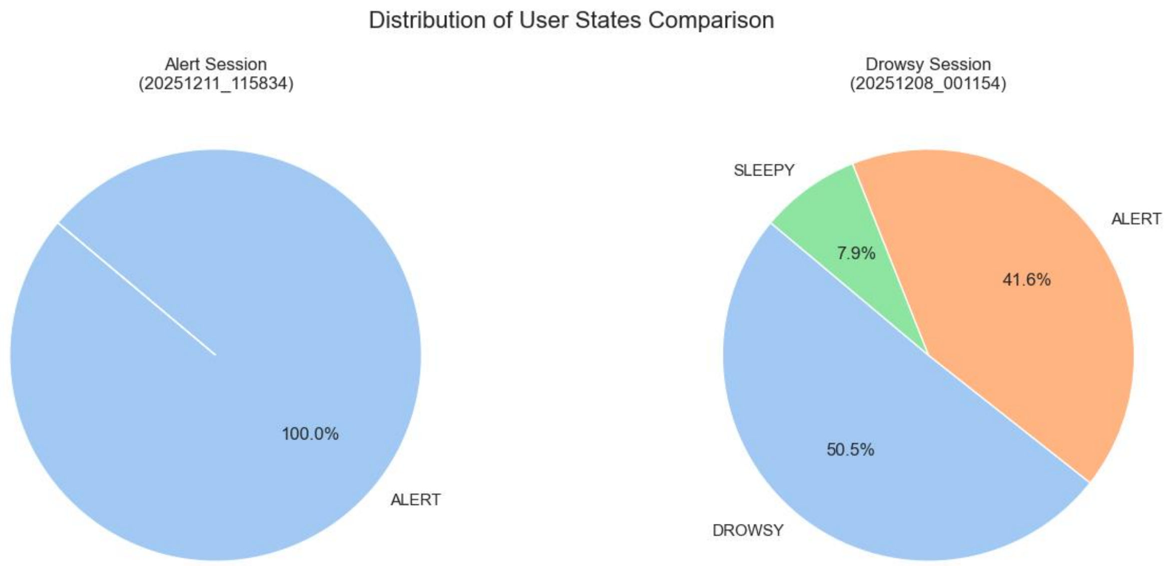
2. Data Collection Procedure

1. A video recording of the participant's face was captured under two conditions: **alert** and **drowsy**.
2. Facial landmarks were tracked for each frame to compute the **Eye Aspect Ratio (EAR)**.
3. PERCLOS values were generated by estimating the percentage of time the eyes appeared partially or fully closed.
4. Reaction-time tests were conducted concurrently, producing a series of timestamped response measurements.
5. All data streams were exported as CSV files for use in Python.

The goal of this stage was to produce synchronized datasets representing physiological and behavioral indicators of alertness.

3. Data Loading and Preprocessing

1. All CSV files were imported using a custom `Dataset` class that handled EAR, PERCLOS, and reaction-time data.
2. Values were coerced to numeric types and cleaned to remove invalid or missing entries.
3. The datasets were aligned by truncating sequences to a common length where necessary.
4. A global **EAR threshold of 22.0** was selected based on statistical inspection of the EAR distributions.
5. Data was prepared for visualization, statistical summaries, and event detection.



	Dataset	EAR/PERCLOS Rows	Reaction Time Rows
0	Alert Session	5517	100
1	Drowsy Session	7052	100

```
Overall comparison of alert vs drowsy sessions:
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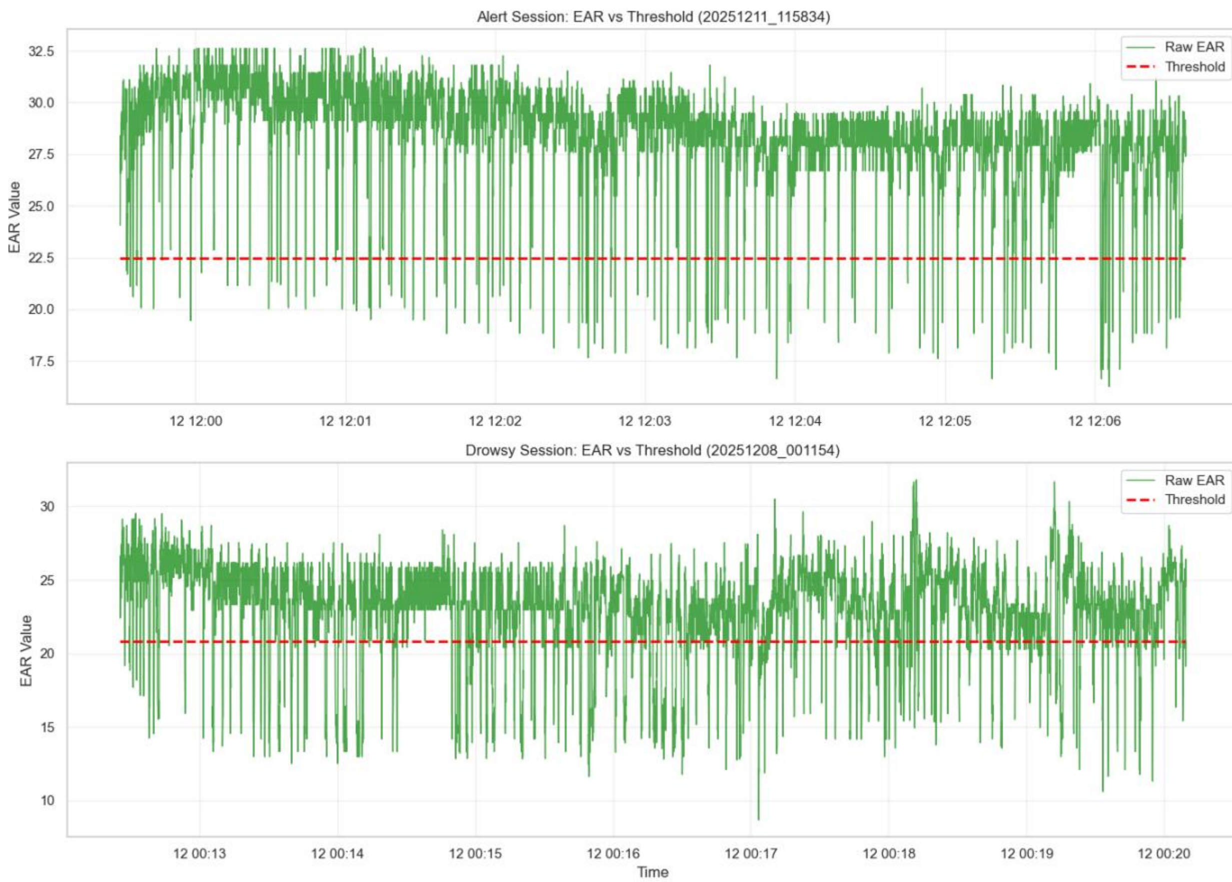
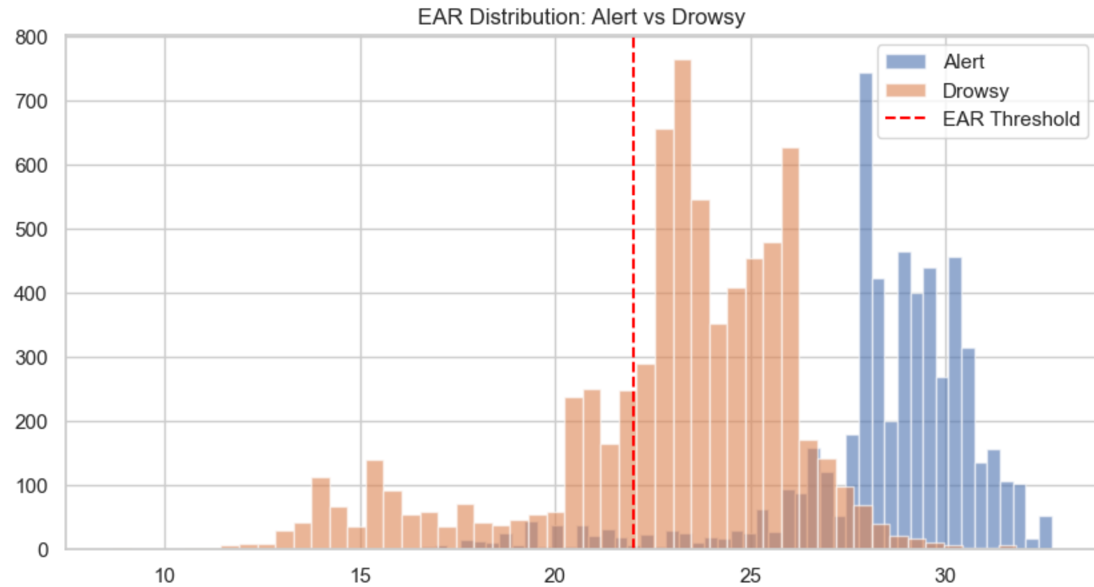
Condition	EAR_mean	EAR_std	PERCLOS_mean	PERCLOS_std	RT_mean_ms	RT_std_ms
Alert	28.39	2.61	1.21	1.97	305.09	37.21
Drowsy	22.94	3.33	17.29	8.83	355.30	117.59

4. EAR Analysis

EAR values were analyzed to evaluate eyelid-openness behavior across both conditions.

- EAR summary statistics (mean, min, max, standard deviation) were computed.
- Distributions were plotted to visualize how eyelid aperture differs between alert and drowsy states.
- Blink events were detected by counting transitions where EAR fell below the defined threshold.





	EAR_drop_events
Condition	
Alert	134
Drowsy	433

	Blink Rate (per min)
Condition	
Alert	43.72
Drowsy	110.52

```
Variability Analysis for EAR and PERCLOS
```

	EAR_std	EAR_range	EAR_MAD	PERCLOS_std	PERCLOS_range	PERCLOS_MAD
Condition						
Alert	2.612	16.45	1.766	1.968	9.3	1.396
Drowsy	3.327	23.14	2.421	8.831	43.0	7.138

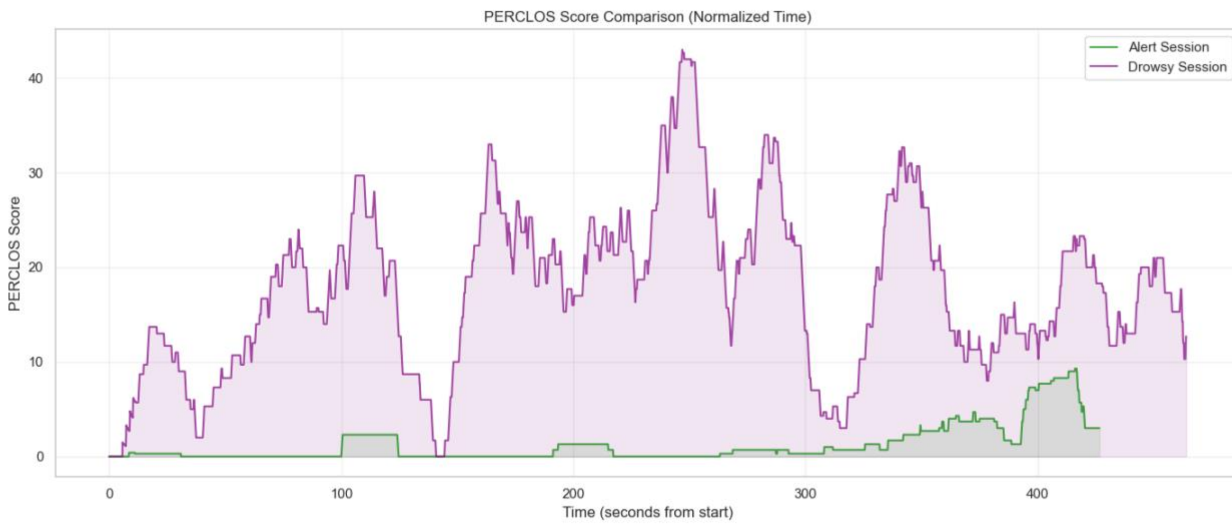
5. PERCLOS Analysis

PERCLOS scores were used to measure sustained eye closure patterns.

Key analyses included:

- Overall distribution of PERCLOS values for alert vs. drowsy sessions
- Identification of periods of high eye closure (fatigue episodes)
- Quantification of total high-PERCLOS time and longest sustained episode

A data-driven threshold (95th percentile) was used because raw PERCLOS values in alert sessions showed extremely low variability.



	StateTransitions
Condition	
Alert	0
Drowsy	34

6. Microsleep-Like Event Detection

Microsleep events were approximated by detecting contiguous periods where PERCLOS remained above the high-drowsiness threshold for at least one second.

Extracted metrics:

- Number of high-drowsiness episodes
- Longest continuous episode
- Total time spent in elevated-fatigue states

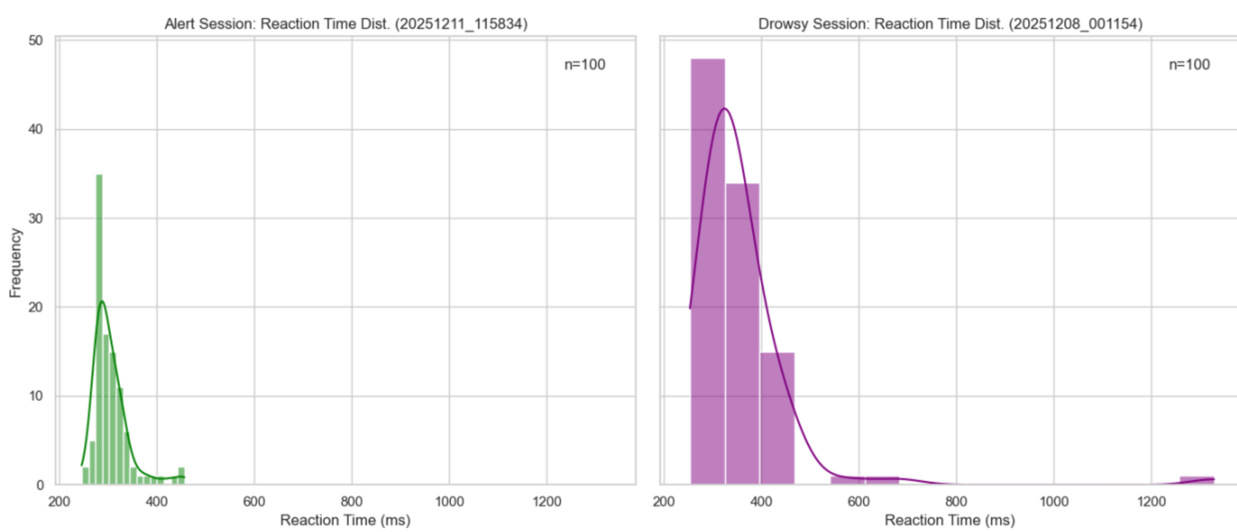
Condition	PERCLOS threshold (95th %)	High-drowsiness events (>=1s)	Total time in high range (s)	Longest high-drowsiness event (s)
Alert	6.0	1	9.567	9.567
Drowsy	32.7	4	12.467	7.700

7. Reaction Time Analysis

Reaction time values were compared between alert and drowsy sessions to assess performance degradation.

Steps included:

1. Aligning reaction-time values with eye-tracking data
2. Calculating average and variability of reaction times across sessions
3. Grouping reaction times by PERCLOS level (low, medium, high)
4. Evaluating whether higher closure levels led to slower responses

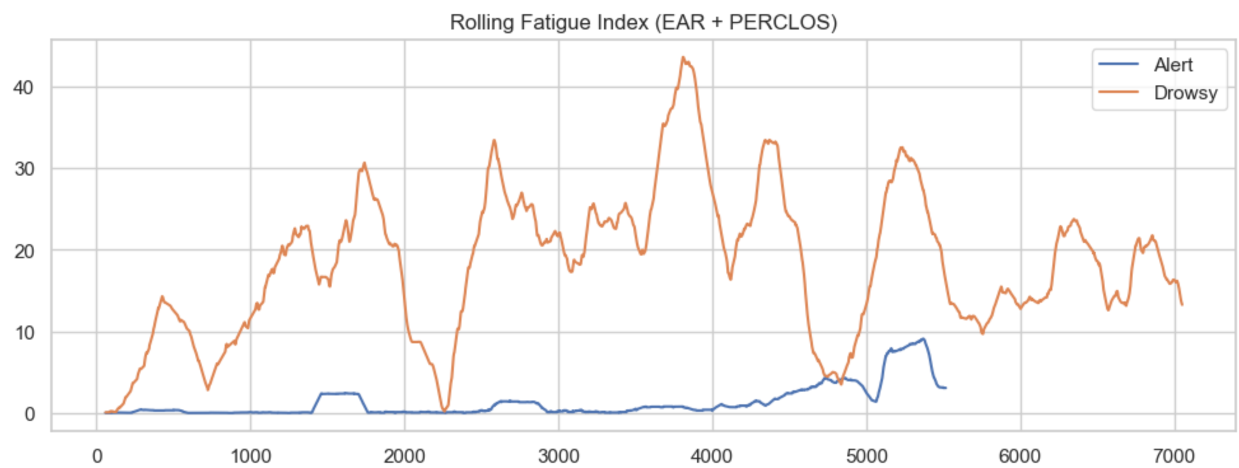


8. Combined Fatigue Metrics

To better capture fatigue trends over time, a rolling fatigue index combining EAR and PERCLOS was computed.

This visualization highlights:

- Gradual buildup of fatigue
- Long-term changes in eyelid behavior
- Differences between alert and drowsy temporal dynamics



9. Summary of Observations

Across all analyses, the drowsy condition demonstrated:

- Lower average EAR values and more frequent dips
- Greater variability in eyelid behavior
- Longer and more frequent high-PERCLOS episodes
- Slower and more variable reaction times
- Higher fatigue index values across time

These results collectively indicate a clear physiological and behavioral separation between alert and drowsy states, supporting the reliability of EAR, PERCLOS, and reaction time as indicators of driver fatigue.

IV. Conclusion

This project successfully demonstrated how EAR, PERCLOS, and reaction time can be combined to detect driver drowsiness and characterize fatigue-related behavior.

What Was Learned

- EAR and PERCLOS provide complementary insight into eye-closure behavior.
- Blink rate increases or becomes irregular as fatigue rises.
- High-PERCLOS periods provide a strong indication of microsleep or severe drowsiness.
- Reaction time tends to slow when fatigue increases.
- Data alignment and preprocessing are critical for accurate analysis.

Challenges Encountered

Data Collection Challenges

- **Lighting consistency:** Small variations in ambient lighting affected eye-tracking accuracy, requiring multiple recording attempts to obtain stable EAR and PERCLOS values.
- **Camera alignment:** Maintaining a fixed camera position and a consistent face angle was difficult, particularly during drowsy sessions where natural head movement increased.
- **Reliable drowsy-state recording:** Capturing authentic drowsiness proved challenging, resulting in variability in eyelid behavior and reaction-time responses.
- **Synchronization of data sources:** Eye-tracking data and reaction-time measurements were collected asynchronously, making it difficult to align the two datasets precisely.

Analysis Challenges

- **Low variability in alert PERCLOS values:** PERCLOS scores were nearly constant during alert sessions, making fixed microsleep thresholds unusable. A data-driven percentile threshold was required.
- **Event detection tuning:** Blink and high-drowsiness event detection required threshold refinement and noise filtering to avoid false detections or missed events.

Suggestions for Improvement

- Collect longer drowsy sessions to better model extreme fatigue.
- Capture synchronized timestamps across all data streams to improve alignment.
- Incorporate head-pose or yawning detection for a more complete multimodal model.
- Train a machine-learning classifier using the extracted features for automated drowsiness prediction.